

MAXIMUM FLOW

Max-Flow Min-Cut Theorem (Ford Fulkerson's Algorithm)

What is Network Flow ?

Flow network is a directed graph $G=(V,E)$ such that each edge has a non-negative capacity $c(u,v) \geq 0$.

Two distinguished vertices exist in G namely :

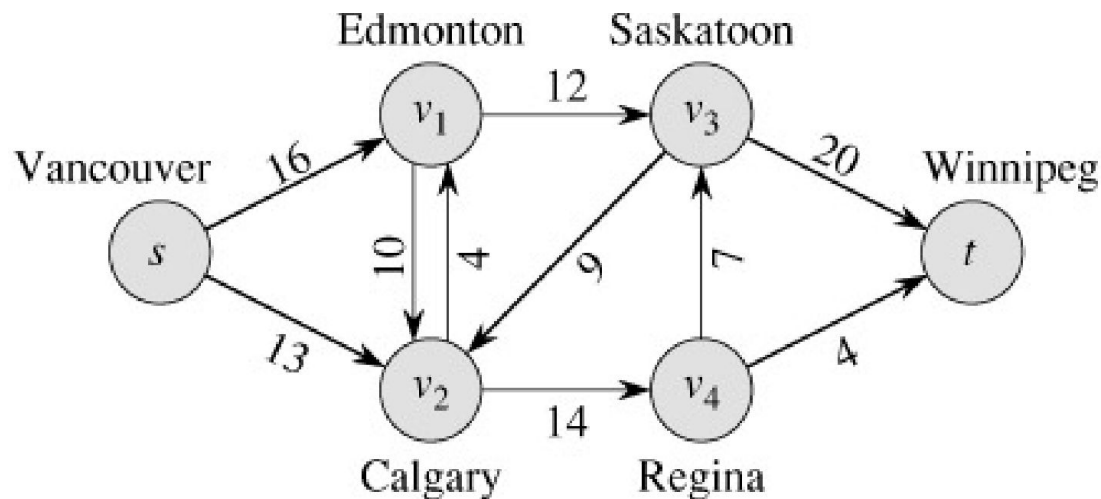
- Source (denoted by s) : In-degree of this vertex is 0.
- Sink (denoted by t) : Out-degree of this vertex is 0.

Flow in a network is an integer-valued function f defined On the edges of G satisfying $0 \leq f(u,v) \leq c(u,v)$, for every Edge (u,v) in E .

What is Network Flow ?

- Each edge (u,v) has a non-negative capacity $c(u,v)$.
- If (u,v) is not in E assume $c(u,v)=0$.
- We have source s and sink t .
- Assume that every vertex v in V is on some path from s to t .

Following is an illustration of a network flow:



$c(s,v_1)=16$
 $c(v_1,s)=0$
 $c(v_2,s)=0 \dots$

Conditions for Network Flow

For each edge (u,v) in E , the flow $f(u,v)$ is a real valued function that must satisfy following 3 conditions :

- **Capacity Constraint :** $\forall u,v \in V, 0 \leq f(u,v) \leq c(u,v)$
- **Flow Conservation:** $\forall u \in V - \{s,t\} \sum_{v \in V} f(v,u) = \sum_{v \in V} f(u,v)$

Skew symmetry condition implies that $f(u,u)=0$.

The Value of a Flow.

The value of a flow is given by :

$$|f| = \sum_{v \in V} f(s, v) = \sum_{v \in V} f(v, t)$$

The flow into the node is same as flow going out from the node and thus the flow is conserved. Also the total amount of flow from source s = total amount of flow into the sink t .

Example of a flow

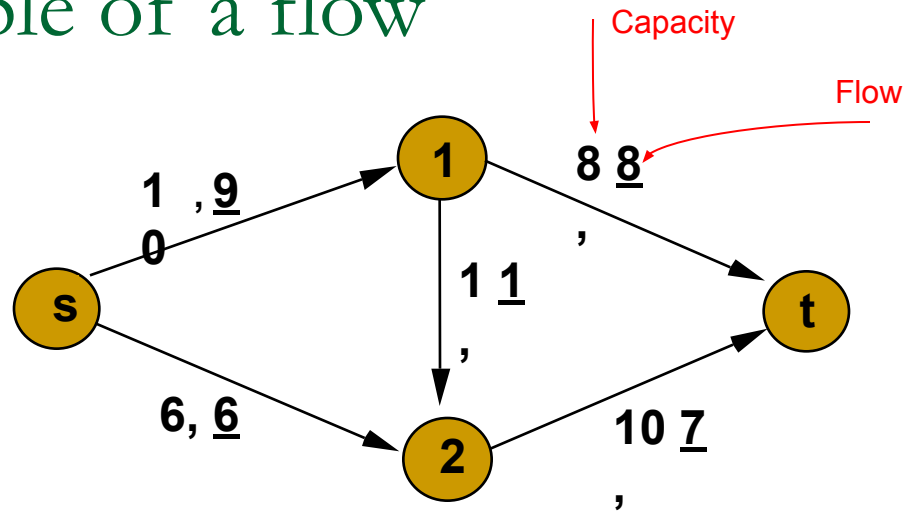


Table illustrating Flows and Capacity across different edges of graph above:

$$f_{s,1} = 9, c_{s,1} = 10 \text{ (Valid flow since } 10 > 9)$$

$$f_{s,2} = 6, c_{s,2} = 6 \text{ (Valid flow since } 6 \geq 6)$$

$$f_{1,2} = 1, c_{1,2} = 1 \text{ (Valid flow since } 1 \geq 1)$$

$$f_{1,t} = 8, c_{1,t} = 8 \text{ (Valid flow since } 8 \geq 8)$$

$$f_{2,t} = 7, c_{2,t} = 10 \text{ (Valid flow since } 10 > 7)$$

The flow across nodes 1 and 2 are also conserved as flow into them = flow out.

The Maximum Flow Problem

Given a Graph $G(V,E)$ such that:

$x_{i,j}$ = flow on edge (i,j)

$u_{i,j}$ = capacity of edge (i,j)

s = source node

t = sink node

In simple terms maximize the s to t flow, while ensuring that the flow is feasible.

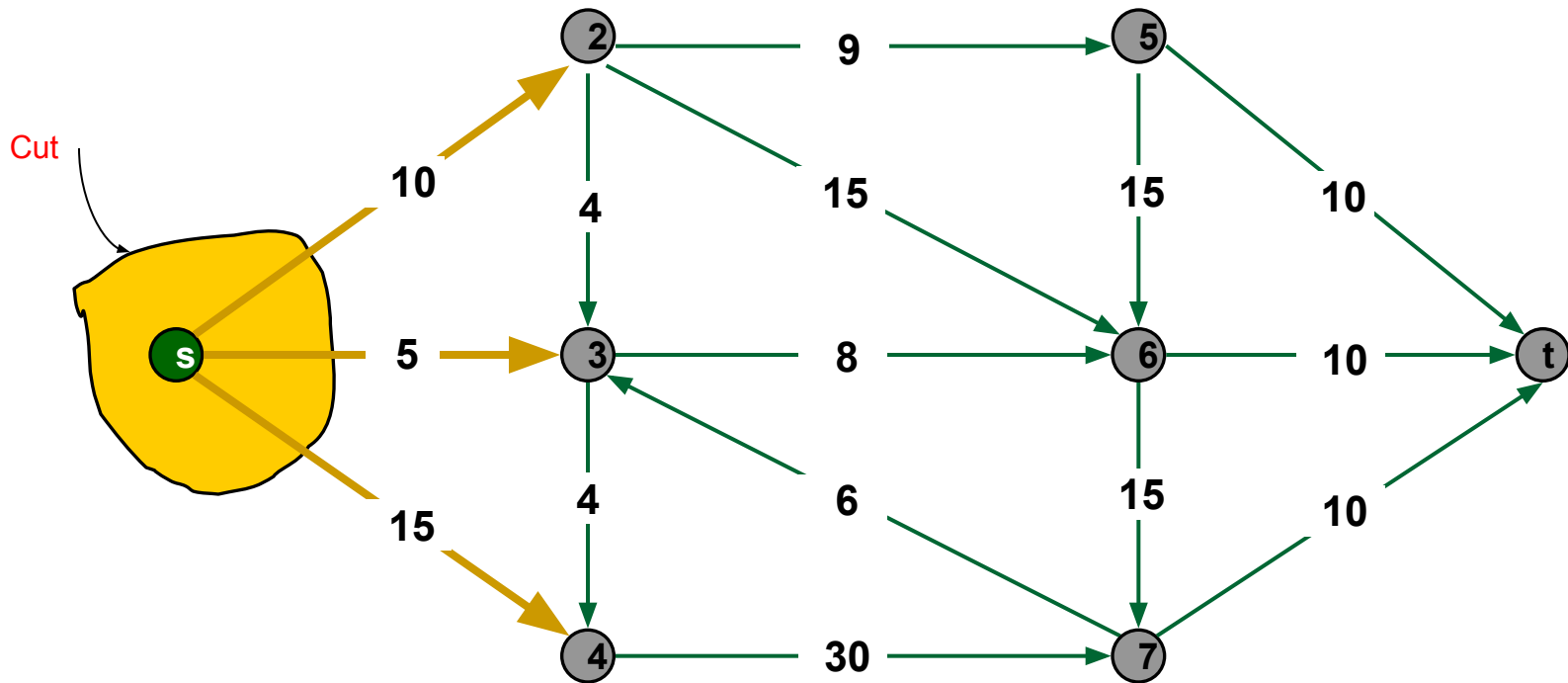
Maximize v

Subject To $\sum_j x_{ij} - \sum_j x_{ji} = 0$ for each $i \neq s, t$

$0 \leq x_{ij} \leq u_{ij}$ for all $(i,j) \in E$.

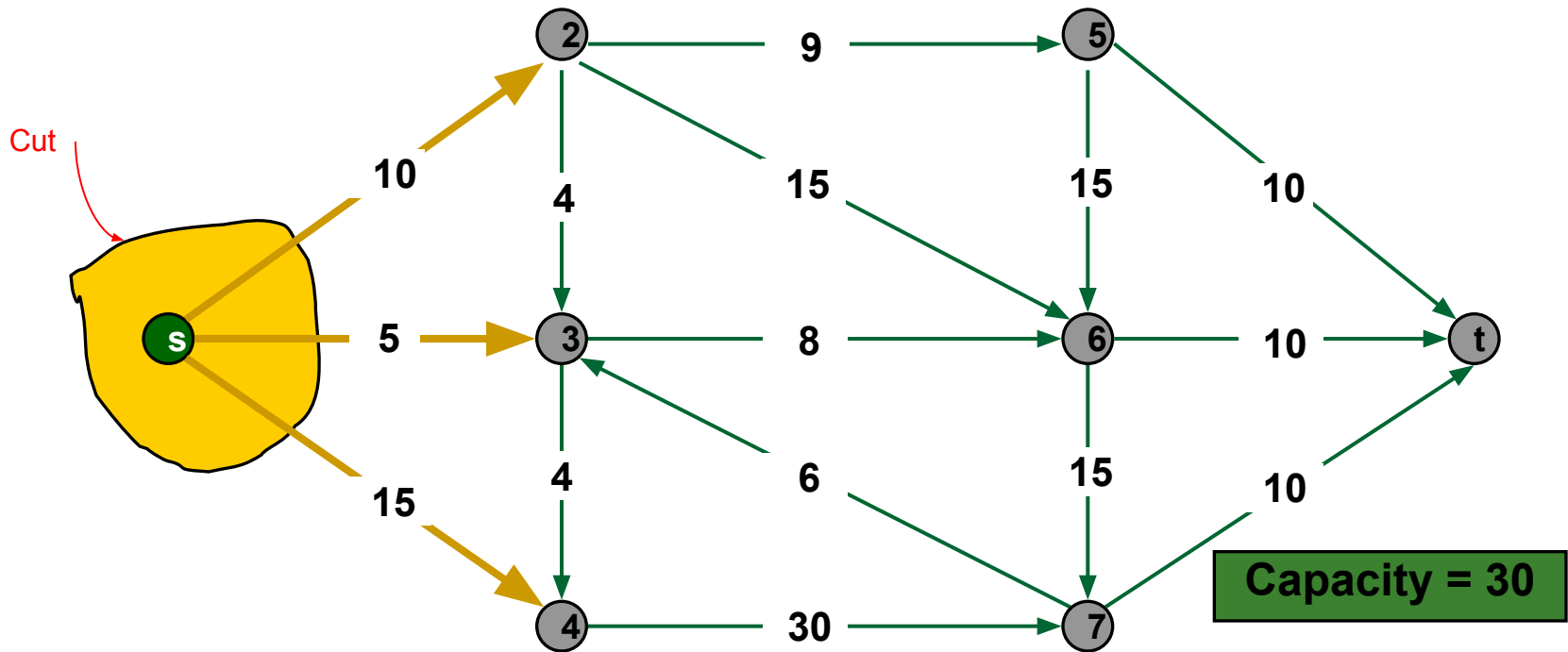
Cuts of Flow Networks

A Cut in a network is a partition of V into S and T ($T=V-S$) such that s (source) is in S and t (target) is in T .



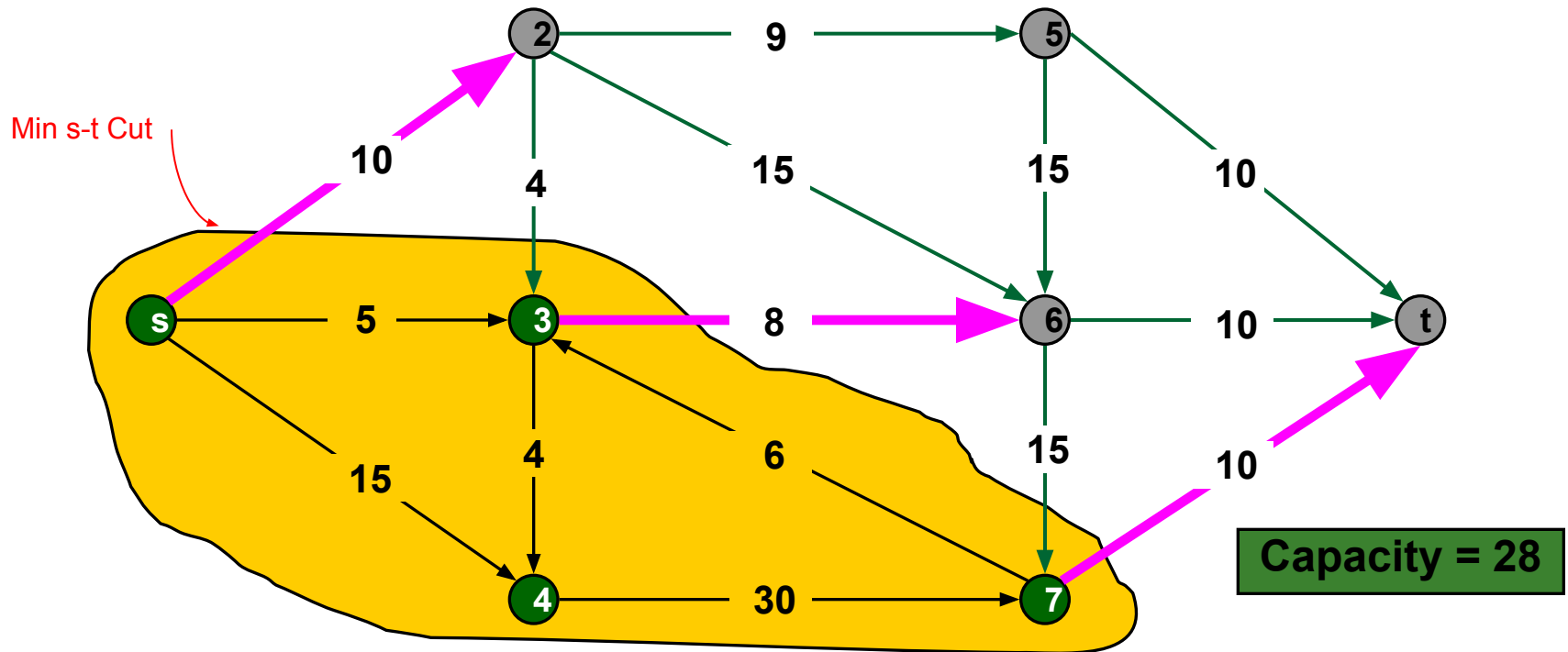
Capacity of Cut (S,T)

$$c(S, T) = \sum_{u \in S, v \in T} c(u, v)$$



Min Cut

Min s-t cut (Also called as a Min Cut) is a cut of minimum capacity



Methods

Max-Flow Min-Cut Theorem

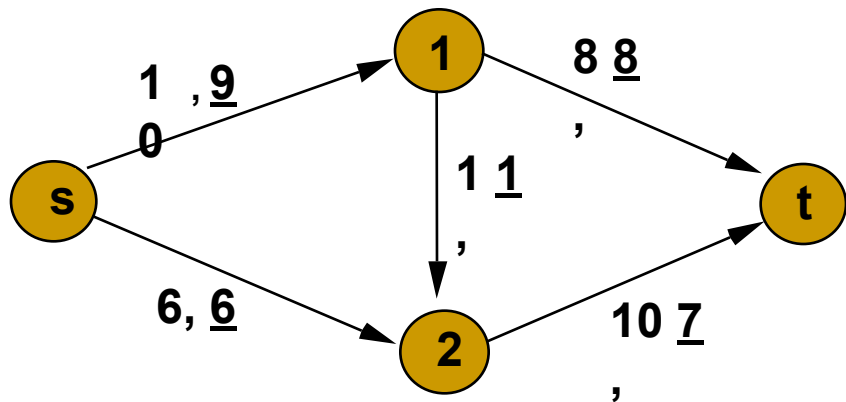
- The Ford-Fulkerson Method
- The Preflow-Push Method

The Ford-Fulkerson Method

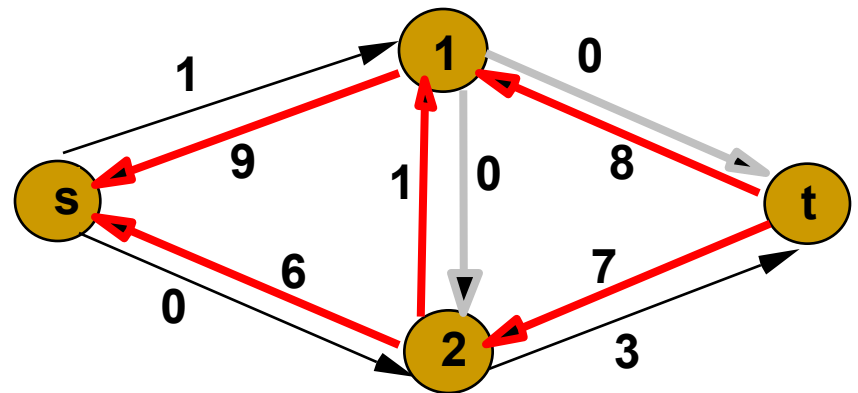
- Try to improve the flow, until we reach the maximum value of the flow
- The residual capacity of the network with a flow f is given by:

The residual capacity (rc) of an edge (i,j) equals $c(i,j) - f(i,j)$ when (i,j) is a forward edge, and equals $f(i,j)$ when (i,j) is a backward edge. Moreover the residual capacity of an edge is always non-negative.

$$c_f(u, v) = c(u, v) - f(u, v)$$



Original Network



Residual Network

-
- *Start with initial flow as 0.*
 - *While there exists an augmenting path from the source to the sink:*
 - *Find an augmenting path*
 - *Determine the amount of flow that can be sent along the augmenting path, which is the minimum residual capacity along the edges of the path.*
 - *Increase the flow along the augmenting path by the determined amount.*
 - *Return the maximum flow.*
-

The Logic of the Residual Network

- The Problem: Once a "pipe" (edge) is partially full, how do we know what's left?
 - The Solution: Build a Residual Graph G_f Components:
 - Forward Edges: $cf(u, v) = c(u, v) - f(u, v)$. (The "Remaining Capacity")
 - Backward Edges: $cf(v, u) = f(u, v)$. (The "Undo Button")
 - Rule: We only move flow through paths that exist in this residual graph.
-

The Ford-Fulkerson Method

Begin

$x := 0$; // x is the flow.

create the residual network $G(x)$;

while there is some directed path from s to t in $G(x)$ do
begin

 let P be a path from s to t in $G(x)$;

$\Delta := \delta(P)$;

send Δ units of flow along P ;

 update the residual network;

end

end {the flow x is now maximum}.

[Click To See Ford Fulkerson's Algorithm In Action \(Animation\)](#)

The Ford-Fulkerson's Algorithm

```
FORDFULKERSON ( $G, E, s, t$ )
```

```
FOREACH  $e \in E$ 
```

```
     $f(e) \leftarrow 0$ 
```

```
 $G_f \leftarrow$  residual graph
```

```
WHILE (there exists augmenting  
path  $P$ )
```

```
     $f \leftarrow$  augment( $f, P$ )
```

```
    update  $G_f$ 
```

```
ENDWHILE
```

```
RETURN  $f$ 
```

```
AUGMENT ( $f, P$ )
```

```
 $b \leftarrow$  bottleneck( $P$ )
```

```
FOREACH  $e \in P$ 
```

```
    IF ( $e \in E$ )
```

```
        // forward arc
```

```
         $f(e) \leftarrow f(e) + b$ 
```

```
    ELSE
```

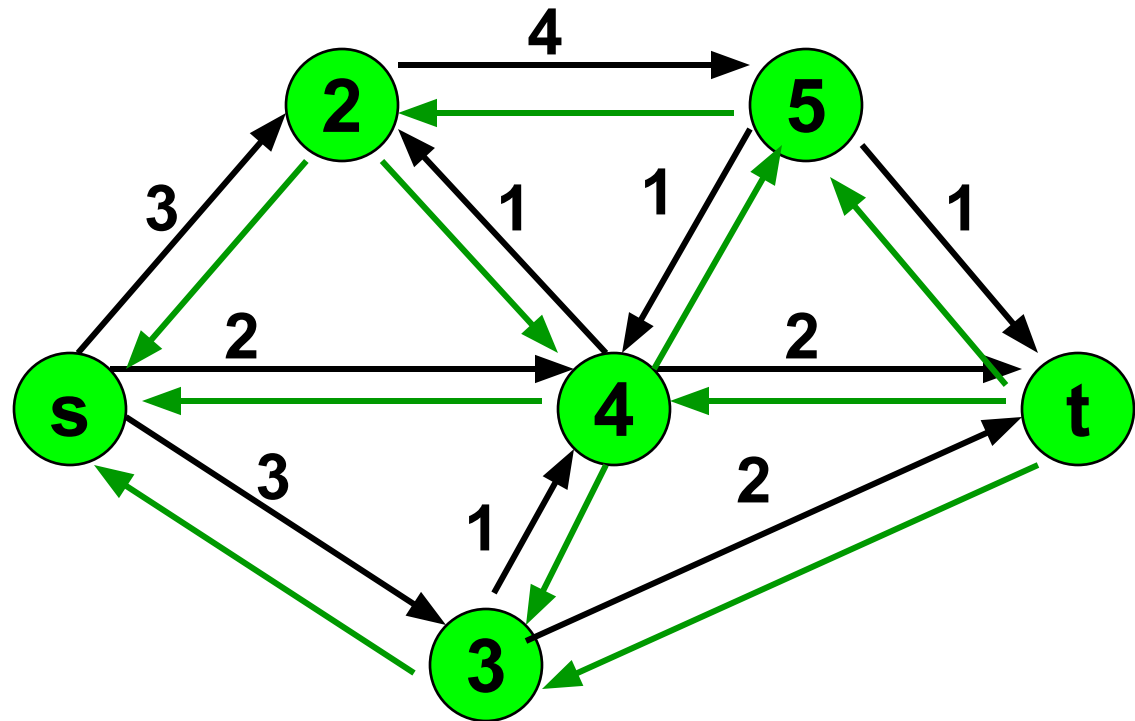
```
        // backward arc
```

```
         $f(e^R) \leftarrow f(e^R) - b$ 
```

```
RETURN  $f$ 
```

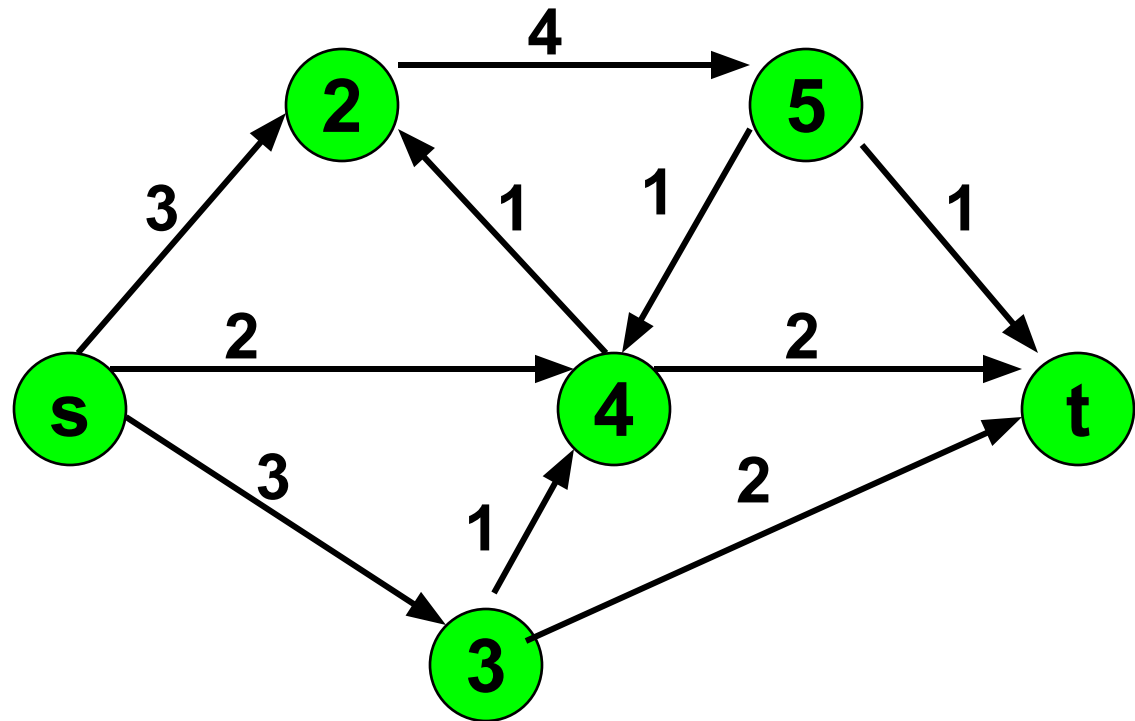
The Ford-Fulkerson Augmenting Path Algorithm for the Maximum Flow Problem

Ford-Fulkerson Max Flow



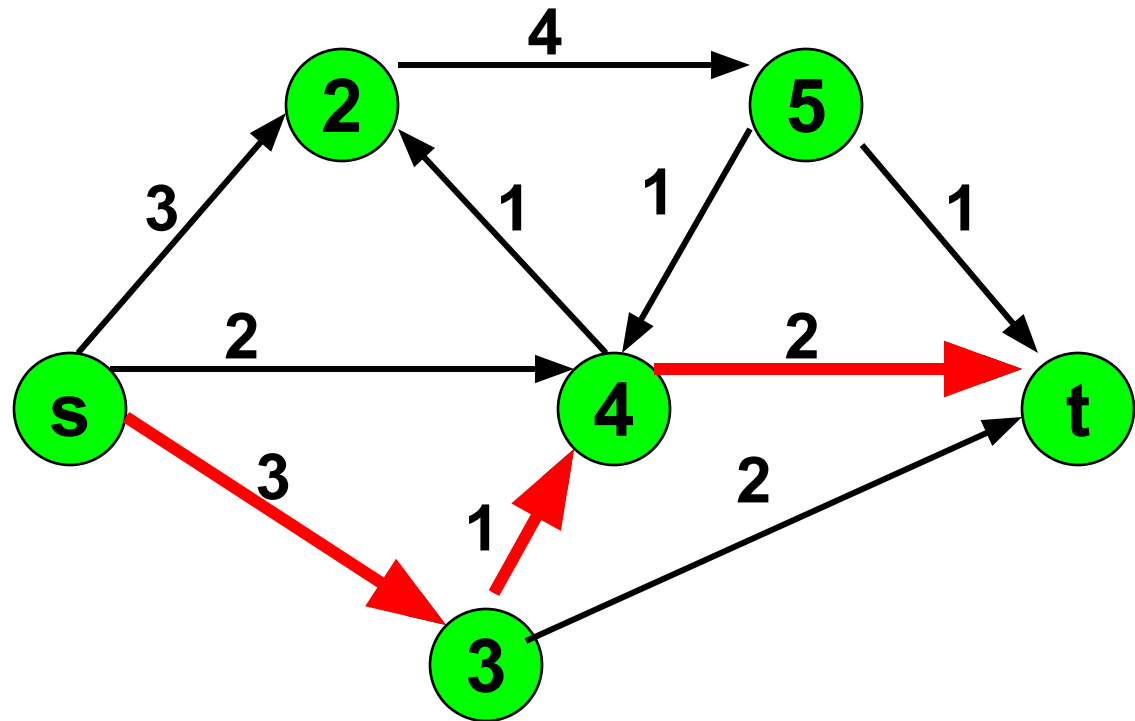
**This is the original network,
plus reversals of the arcs.**

Ford-Fulkerson Max Flow



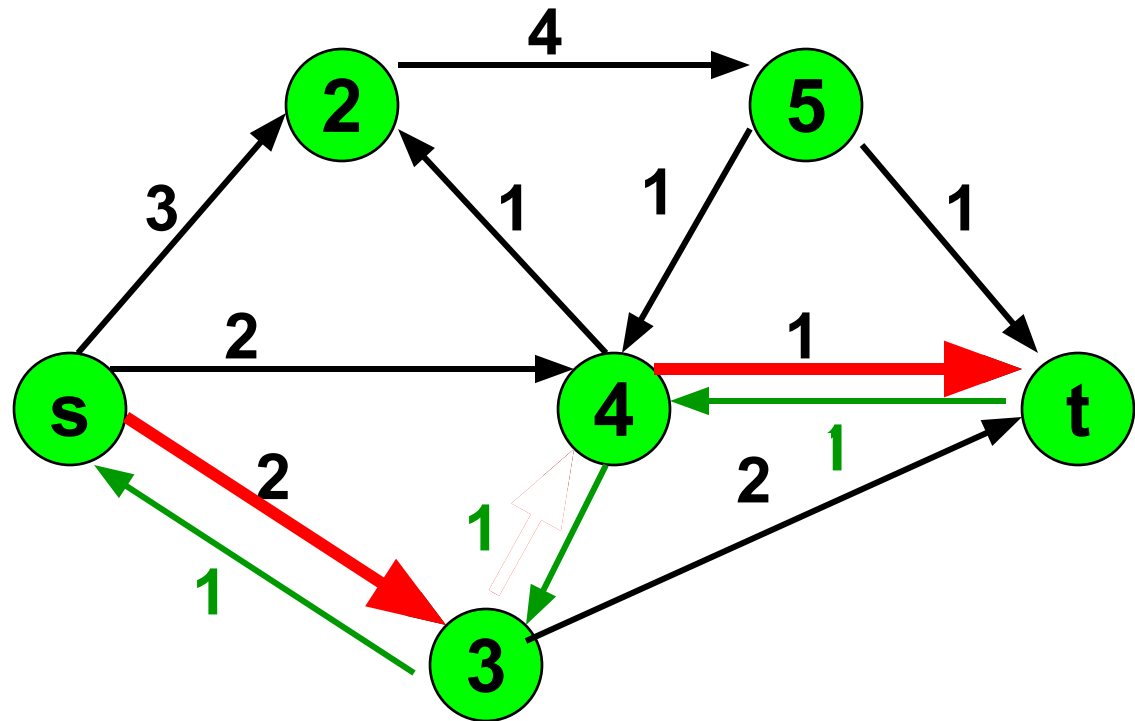
**This is the original network,
and the original residual
network.**

Ford-Fulkerson Max Flow



Find any s-t path in $G(x)$

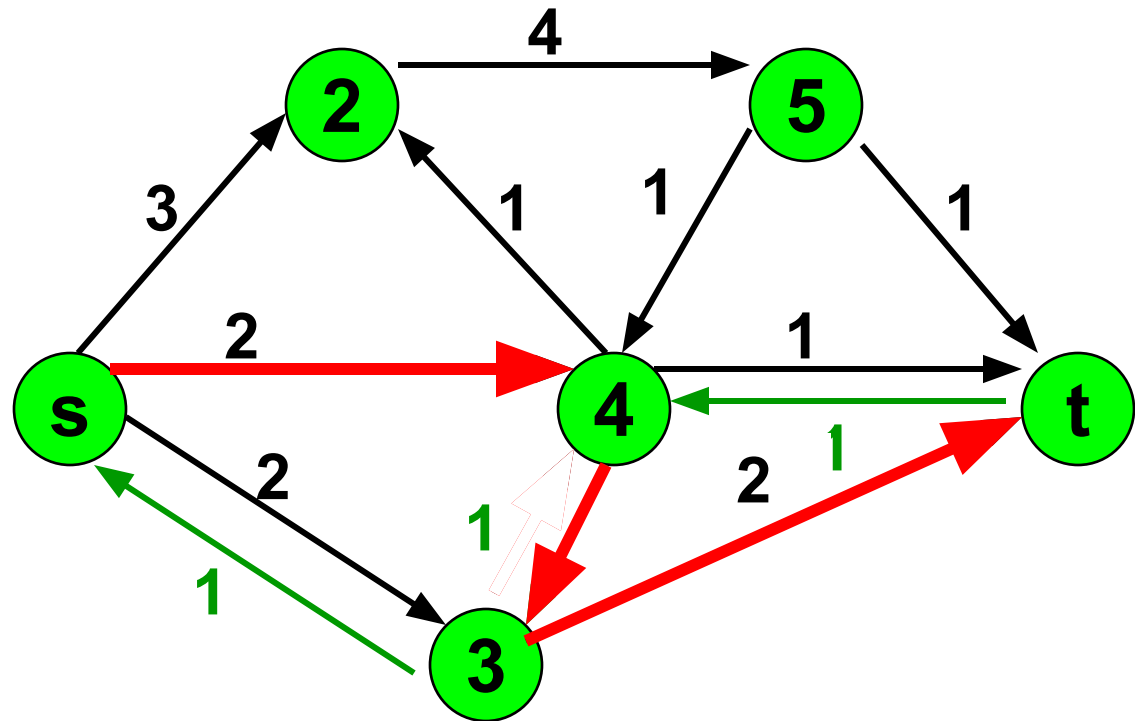
Ford-Fulkerson Max Flow



Determine the capacity Δ of the path.

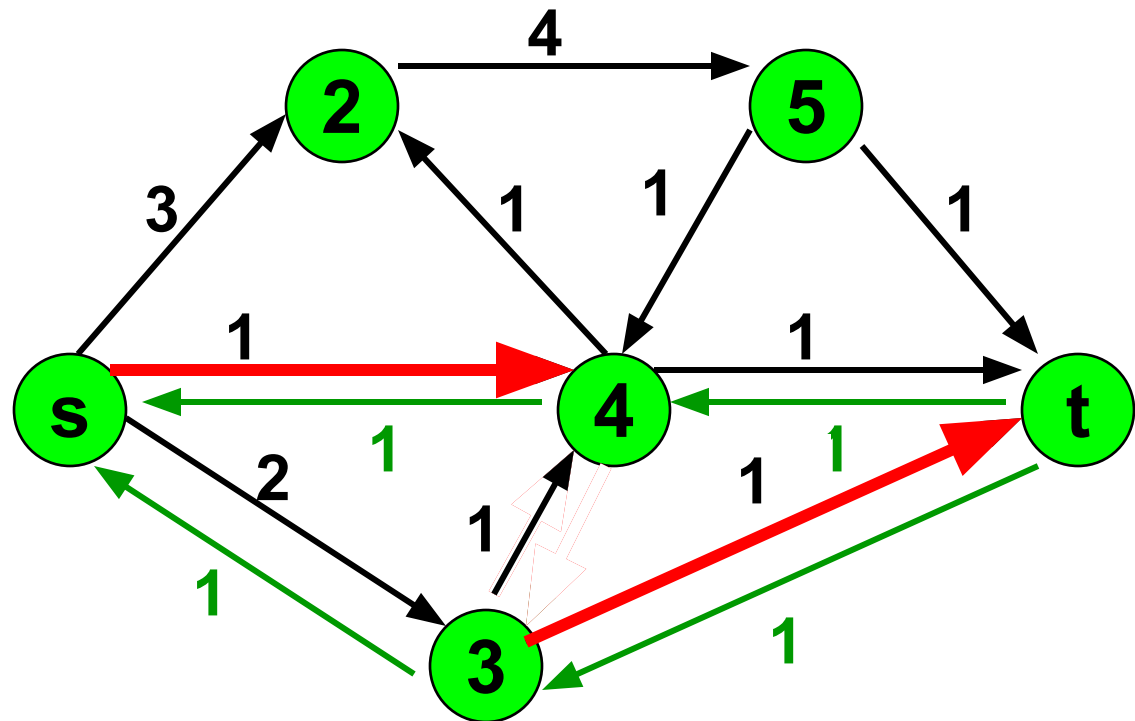
**Send Δ units of flow in the path.
Update residual capacities.**

Ford-Fulkerson Max Flow



Find any s-t path

Ford-Fulkerson Max Flow

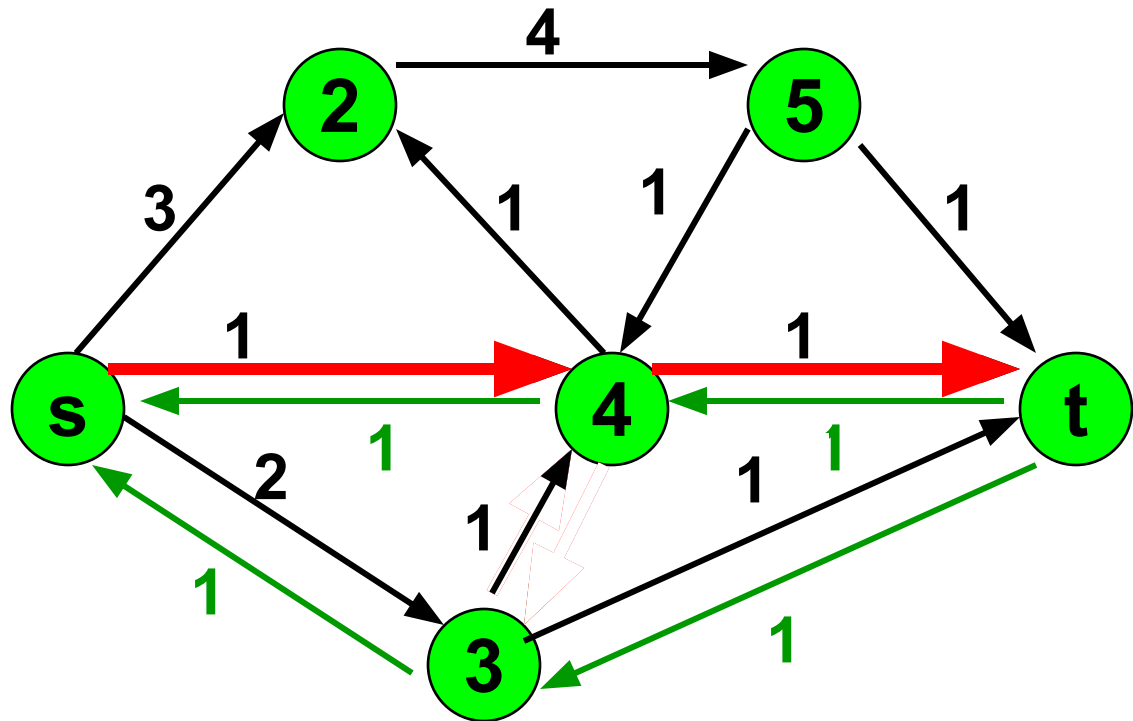


Determine the capacity Δ of the path.

Send Δ units of flow in the path.

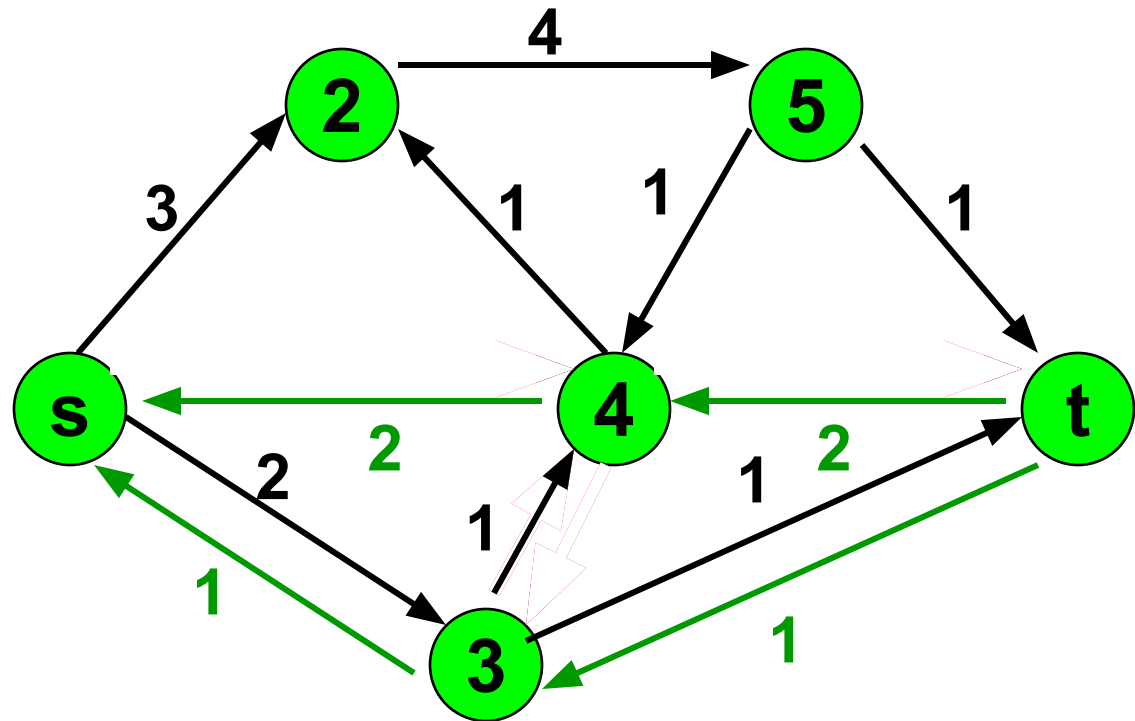
Update residual capacities.

Ford-Fulkerson Max Flow



Find any s-t path

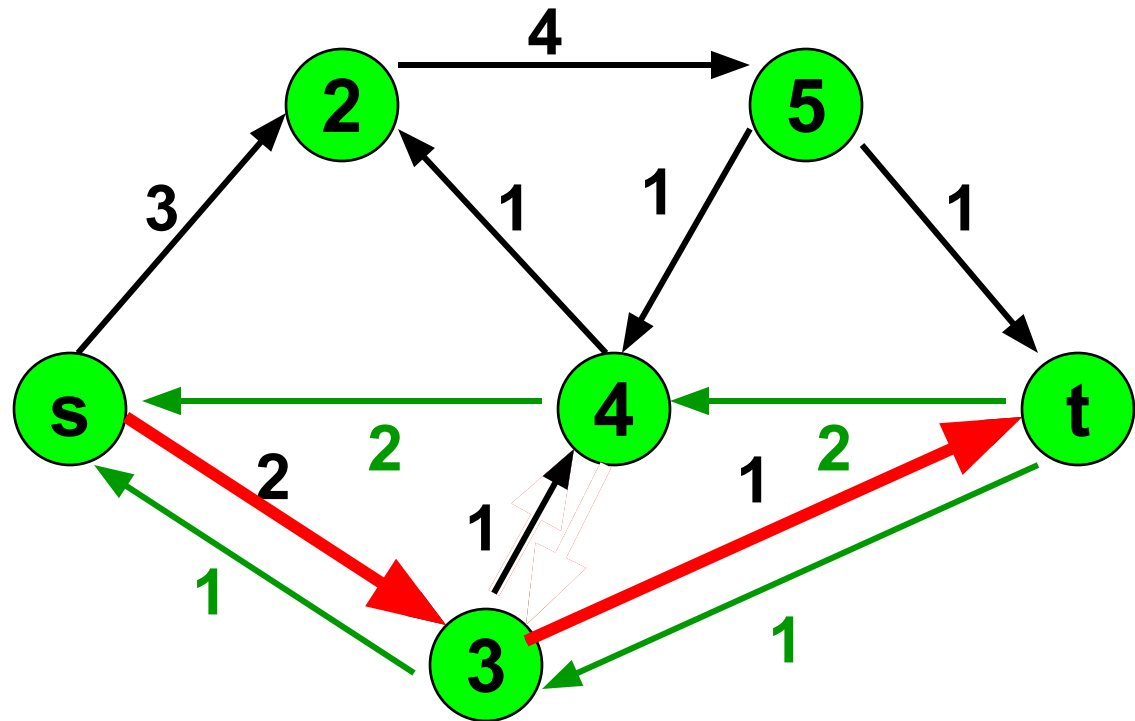
Ford-Fulkerson Max Flow



Determine the capacity Δ of the path.

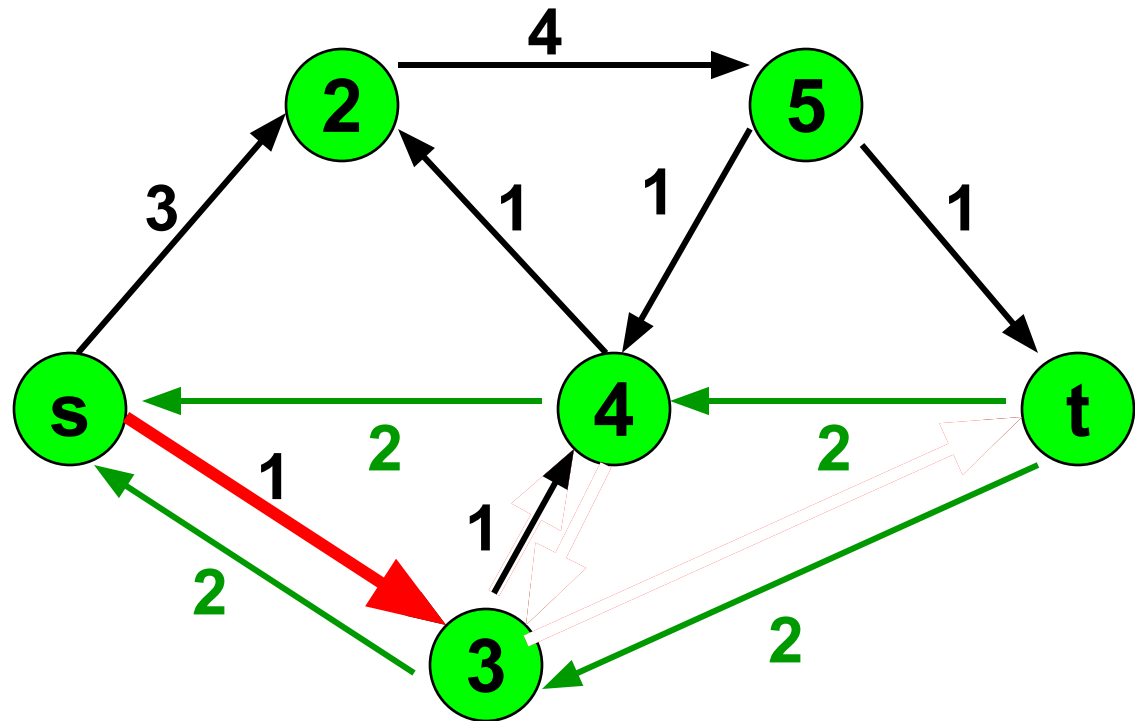
**Send Δ units of flow in the path.
Update residual capacities.**

Ford-Fulkerson Max Flow



Find any s-t path

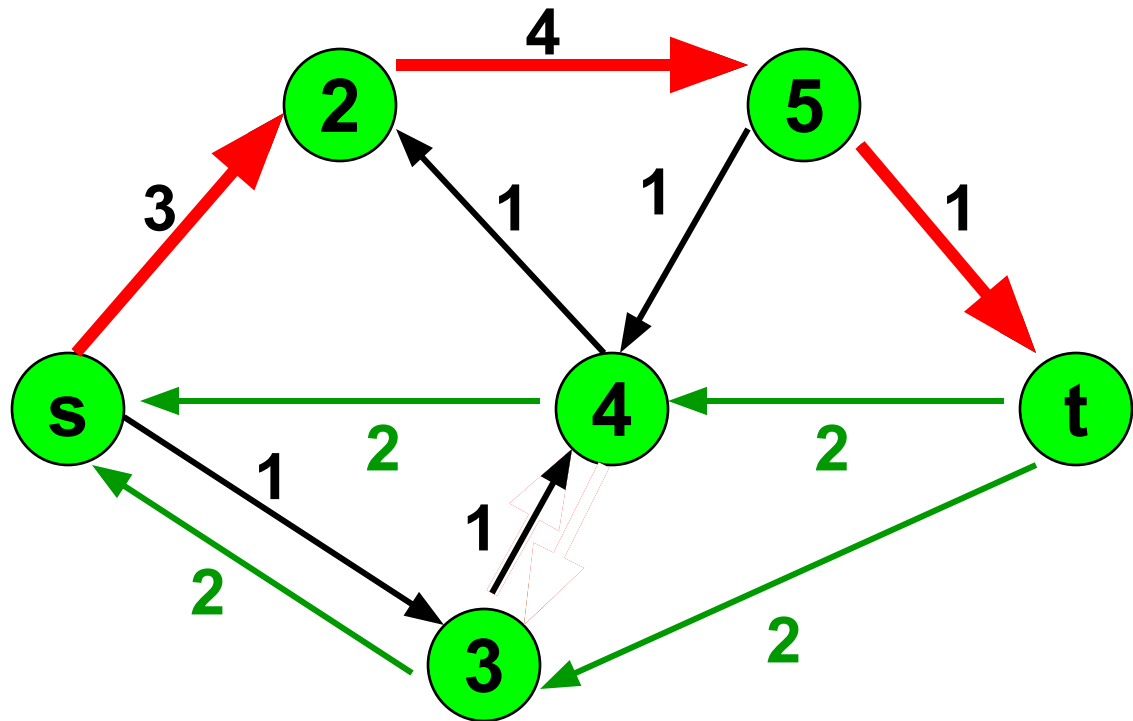
Ford-Fulkerson Max Flow



Determine the capacity Δ of the path.

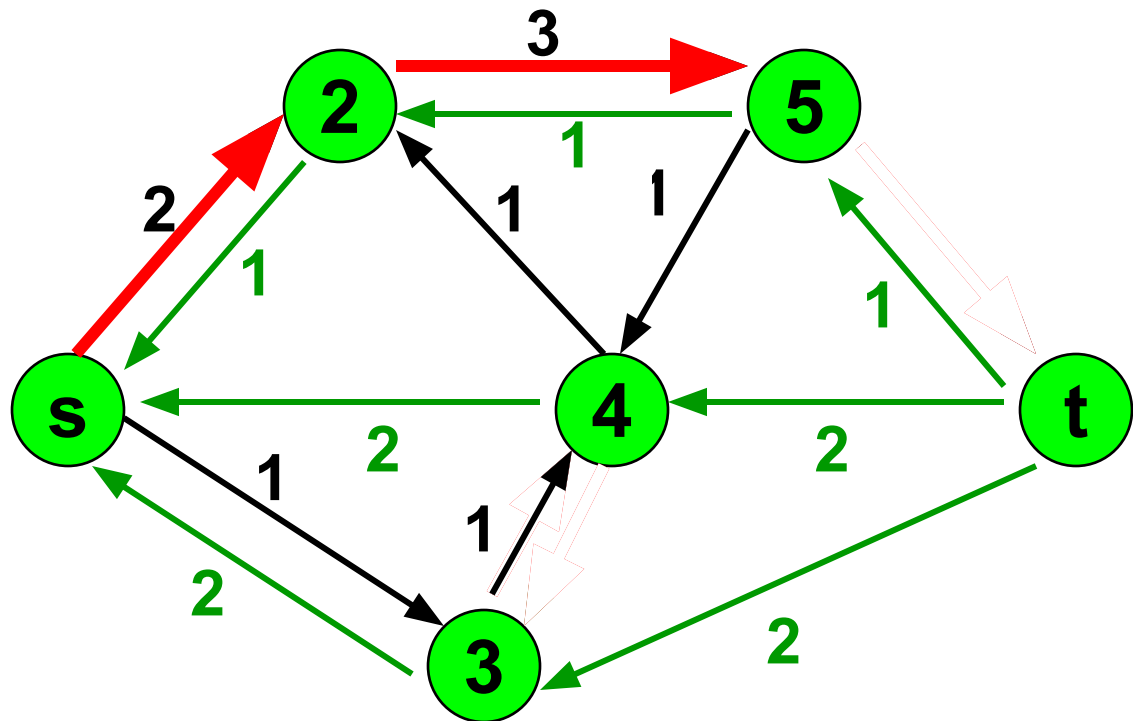
**Send Δ units of flow in the path.
Update residual capacities.**

Ford-Fulkerson Max Flow



Find any s-t path

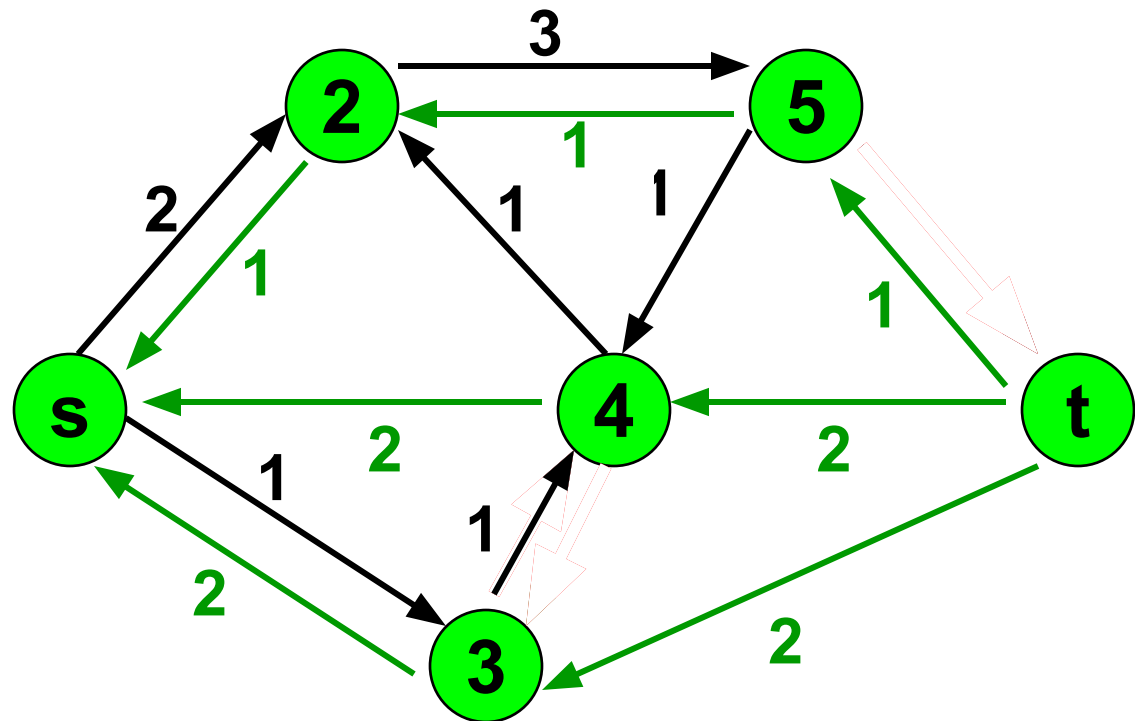
Ford-Fulkerson Max Flow



Determine the capacity Δ of the path.

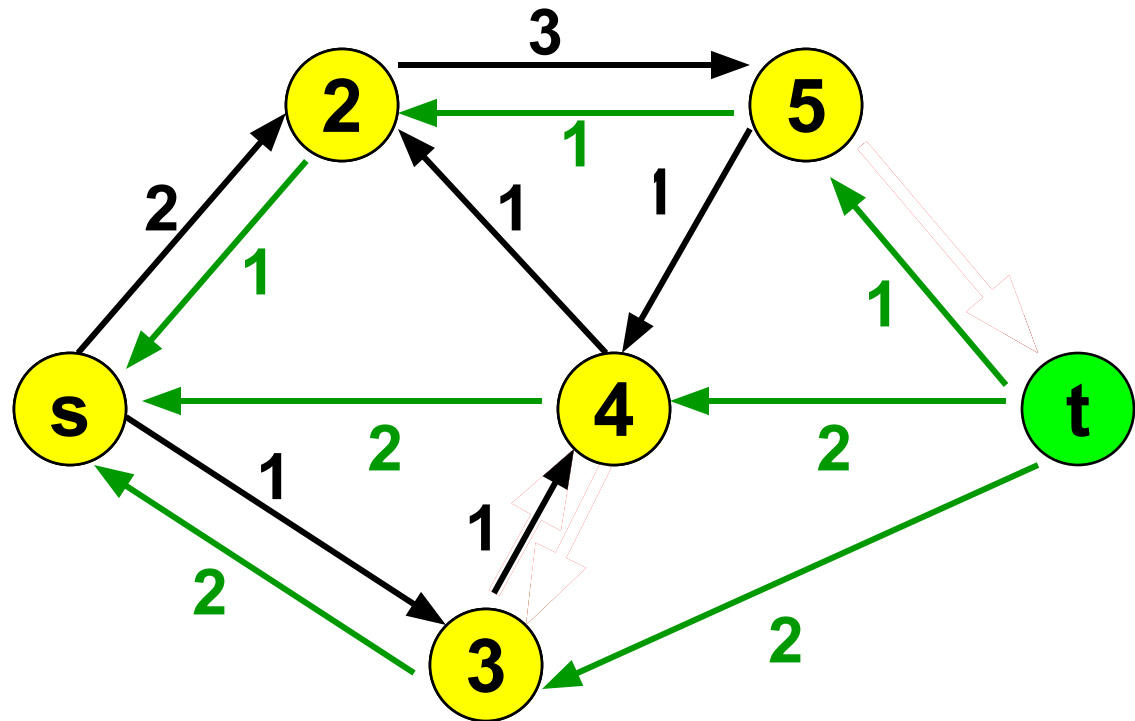
**Send Δ units of flow in the path.
Update residual capacities.**

Ford-Fulkerson Max Flow



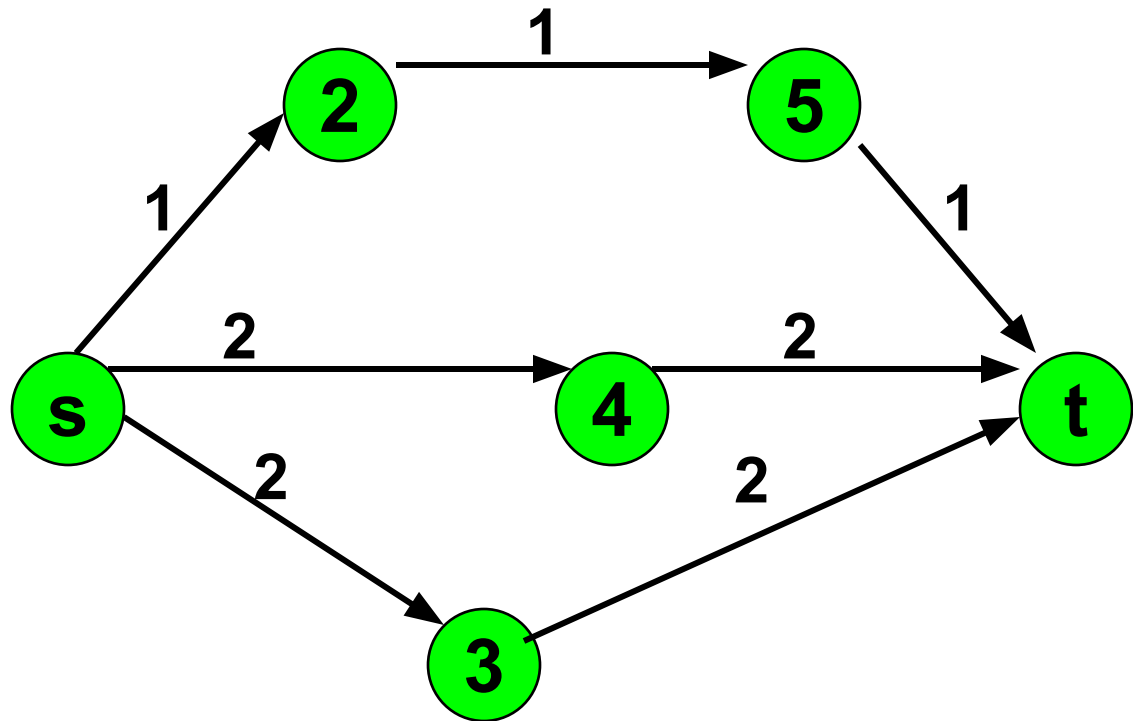
There is no s-t path in the residual network. This flow is optimal

Ford-Fulkerson Max Flow



These are the nodes that are reachable from node s.

Ford-Fulkerson Max Flow



Here is the optimal flow